Ricardo Rodrigues da Silveira Filho

Biologist (PhD)
Data Scientist
Developer

CONTACT

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SKILLS

Languages
TypeScript, Python, R, SQL (PL/pgSQL), HTML, CSS, Shell

Backend
Node.js, FastAPI, Prisma, PostgreSQL, Supabase, Redis, PowerSync

Frontend and mobile
React, Next.js, Tailwind, PWA, Capacitor (Android), Shiny

Infrastructure and testing
Git, Docker, nginx, GitHub Actions, Cloudflare, Vitest, testthat

Data and analytics
ecological statistics, geospatial data (QGIS, PostGIS), Power BI (DAX, Power Query), ggplot2, Quarto and R Markdown, emissions inventory (GHG Protocol)

Applied AI
agent-assisted development (Claude Code), chatbots (Blip), team AI adoption

Product and management
Product Owner, requirements gathering, ADRs, Scrum, project management (PMI), pre-sales and R&D grant proposals

LANGUAGES

Portuguese (native)
English (fluent)
Spanish (basic)

Summary

Data scientist and software developer with a PhD in Zoology, two postdocs and more than fifteen years in environmental consulting. I know the data from the moment it's collected, and I build the system that receives it. For the last two years I led the Innovation area of an environmental consultancy as Product Owner and Tech Lead, with internal systems used across the company, an ESG management SaaS platform, a field data collection suite and BI dashboards for clients. Before that, I built and ran a customer service chatbot with thousands of monthly users and helped create a delivery app for neighbourhood shops. I code in TypeScript, Python and R, lead teams with clear scope and documented decisions, and use generative AI in development, with the gain measured in hours.

Experience

Seteg 2024–present

Innovation Analyst II (Jun 2025 – present) · *Environmental Analyst* (Nov 2024 – Jun 2025) · environmental and geological consultancy

- As Product Owner, conceived and led the project and time management system that replaced a paid tool and is used by the **whole company (about 45 people)**, with a closed beta as the go-live gate.
- Led the automation of processes across the **whole company**: the team built **12 internal systems**, one for each department, and a **single portal** that brings them together with **5 integrated platforms** in one place, along with policies and support channels. They include 8 request systems used by **about 40 people in 5 departments**; I handled requirements gathering, delivery acceptance and the permission matrix.
- Designed the data model of a field data collection suite (PostgreSQL/PostGIS, Supabase with RLS, offline-first sync) and led the company's first project management system (Fastify, Prisma, React), with ERP integration and 189 tests, reviewing the code and coding alongside the team when needed.
- Developed the goals module of an ESG management SaaS platform funded by an innovation grant, and drove the final deliveries through to production deployment.
- Coordinated a team of 4 (3 developers and 1 designer), with direct management and an individual development plan for one developer; Innovation's point of contact with the board and the PMO.
- Wrote the area's **operating manual**: how requirements are written, the git workflow, how project boards are kept and the delivery cycle, from request to handover.
- Measured the productivity gain from generative AI in three departments (**about 70% less time**, based on logged hours) and wrote the company's first AI usage policy.
- Delivered **GHG Protocol** emissions inventories, in the ESG platform and in emissions and target dashboards (Power BI, DAX), and a hydropower generation analytics solution for an energy sector client (FastAPI, trend tests).
- Built tools for field work and management: a **field photo metadata extractor** (EXIF, OCR and reverse geocoding, shipped as a single executable), a forest inventory progress dashboard for a client (a static site reading the field spreadsheet on SharePoint through Microsoft Graph) and a weekly team hours dashboard.

Conciso TI 2020–2024

Data Scientist (Feb 2020 – Oct 2024) · technology

- Conceived, built and ran **Nubus**, the company's customer service chatbot on the Blip platform: **3,000 to 6,000 unique users a month** (peak of 10,000) and a rating always above 4.5 out of 5 (usually **4.7 to 4.8**). I was the point of contact with Blip and automated the **monthly usage reports** in R (ggplot2, R Markdown).
- Built the **monthly demand forecast** for customer service: time series modelling with **Prophet in R** to estimate how many people would reach the chatbot the following month. Customer service, sales and the board used it to prepare for the month ahead.
- Co-conceived **Na Esquina** with the two other developers on the team: a delivery app for small and mid-sized grocery stores, built in six months during the pandemic, before the major delivery apps served neighbourhood shops. I created the name and the brand and built the Firebase database, with my own pipeline that listed products and pulled their images from the web to build the catalog.

Elementus Soluções Ambientais 2023–2025

Senior Project Analyst (Oct 2023 – Apr 2025) · environmental consultancy

- Technical lead (registered with the Biology Council) for sea turtle monitoring at the Port of Natal: quarterly boat surveys, georeferenced records and technical reports to meet the licence conditions.

Academic research 2010–2023

UFRN/PPBio · UNEMAT · UFPB

- *Postdoc, UFRN/PPBio* (2022–2023): pipeline for Brazilian vertebrate checklists and genetic data coverage (BOLD, GenBank), which grew into tetrapoDNA.
- **Collaborating researcher at the Knowledge Center for Biodiversity** (Brazilian biodiversity research network led by UFMG) since 2022, after my PPBio postdoc. Through the network: two letters in *Science* (2023 and 2024) and a paper in *Perspectives in Ecology and Conservation* (2026).
- *Postdoc, UNEMAT* (2018–2019): woody community dynamics in the Cerrado; co-supervision, thesis committees and R courses.
- *MSc and PhD in Zoology, UFPB* (2010–2017): mimicry and phylogeography of snakes, including a one-month research visit to the Natural History Museum, London (2011), surveying its coral snake collection with the museum's researchers.
- **Assessor for the Pernambuco state Red List of amphibians and reptiles** (2014 workshop, IUCN criteria, snakes group) and co-author of the state's official lists of threatened species (amphibians, 2015; reptiles, 2017).
- **15 peer-reviewed papers and 1 accepted (2026)**, including two letters in *Science* (2023 and 2024).

Freelance environmental and statistical consultant 2010–present

- Herpetofauna work for environmental licensing in energy, mining and infrastructure, hired by consultancies (ADL, Life, Maron, Ophian, Seteg, Spizaetus, Táxon): monitoring programmes and baseline surveys for impact assessments and licensing reports at wind and solar complexes (**ENEL, ENGIE, AES Brasil, Elera**), a bauxite mine in the Amazon (**Mineração Rio do Norte**), a railway (Transnordestina), a thermal power plant (CELSE) and a gas pipeline (TAG).
- Ecological analyses for herpetofauna monitoring in environmental licensing (richness, diversity, similarity, NMDS, rarefaction), with didactic reports in R and Quarto (2023–present).
- Data analysis for the **impact assessment reports on how the Manati gas pipeline and platform affect small-scale fisheries in Bahia, delivered to Petrobras** (1st to 4th report, 2023–present): catch, CPUE, cost and income statistics by fishery in the 1st report; for the 4th, a **20-year landing database (over 1 million records**: about 366,000 trips and 700,000 species records) rebuilt in R from raw spreadsheets and a legacy Python database, an audit and correction of the historical series, trend tests (Mann-Kendall, Sen) by fishing ground and distance to the project, and maps and tables generated by the pipeline.
- Co-founder of **Consulte um Cientista** (2020–2023), a statistics and research consultancy for companies and individuals, with analyses in R and didactic Quarto reports.

Barbosa & Barbosa 2008–2009

CAD Drafter (2008 – 2009)

- Technical drawings for architectural projects in AutoCAD.

Projects

- **tetrapoDNA and traqueophyDNA.** Public dashboard showing which Brazilian tetrapods (about 4,800 amphibian, reptile, bird and mammal species) still have no DNA sequence in BOLD or GenBank. R pipeline with synonym checks (CTFB, GBIF) to avoid false gaps, weekly updates via GitHub Actions and a GitHub Pages site. *traqueophyDNA*, under construction, applies the same method to the roughly 35,000 native vascular plants in Flora e Funga do Brasil, with synonyms from WCVP and the NCBI taxonomy.
  rsilveirafh.github.io/tetrapoDNA
- **claude-plugins.** Open-source catalog of my Claude Code plugins. *banca-simulada* reviews research proposals, from undergraduate theses to faculty hiring, the way a committee would: it points to the passage, the severity and a suggestion across eight axes, and lists the likely questions. *memoria-skills* turns a Markdown vault into a memory system with human review.
  github.com/rsilveirafh/claude-plugins

- **R packages for species checklists (amphiBR and ReptiliaBR).** Open-source contributions: the 2026 Brazilian amphibian checklist (1,251 species) and threat status from the 2026 national red list.
- **Online stores and custom websites.** Two confectionery stores on the same template (Next.js, Supabase, PIX instant payments, PWA), deployed and tested in production: product configurator, server-side pricing, a real-time order panel and 259 automated tests. I also build websites for individuals, businesses and research labs from two templates of my own, in one or two languages.
  rsilveirafh.github.io/modelos-de-site
- **Carona Solidária.** Ride-sharing platform for voters with reduced mobility in the 2026 runoff election, in three state capitals.
- **rr-quant.** Personal quant dashboard that runs itself every trading session (Python, Vercel): a pre-market snapshot of 46 assets, regime and correlation reads and a logistic-regression scoreboard for the Ibovespa.
  github.com/rsilveirafh/rr-quant

Education

Postdoc, Conservation, UFRN / PPBio	2022–2023
Postdoc, Plant ecology, Mato Grosso State University (UNEMAT)	2018–2019
PhD in Zoology, Federal University of Paraíba (UFPB)	2017
MSc in Zoology, Federal University of Paraíba (UFPB)	2012
BSc in Biological Sciences (teaching), State University of Feira de Santana (UEFS)	2009